CHEM F111 : General Chemistry

Semester II: AY 2023-24

Lecture- 16, 21st February 2024, Wednesday

innovate achieve lead

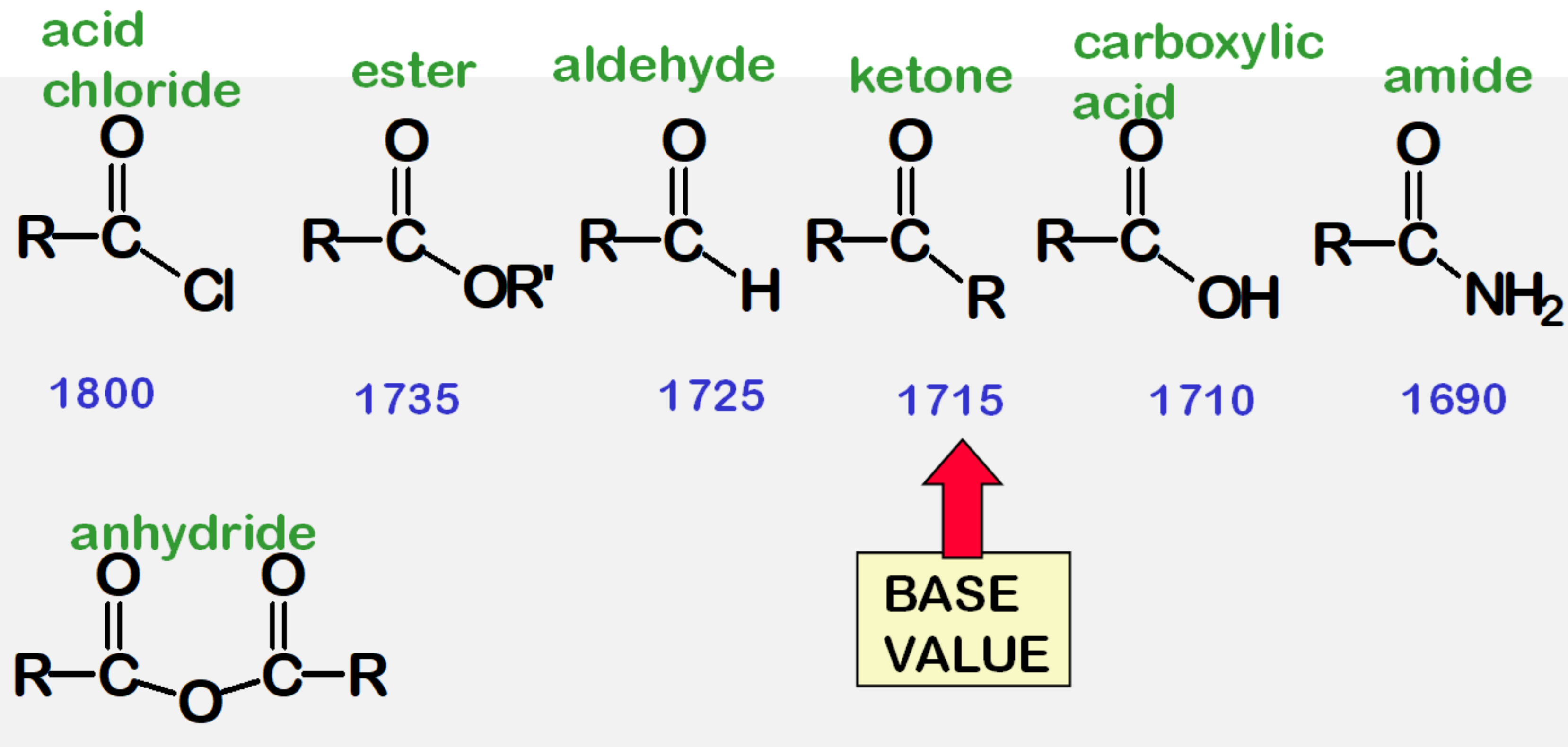
Last Class:

- Degree of freedom; Vibrational modes in polyatomic molecules - normal modes; superposition of normal modes;
- Normal modes of vibrations of carbon dioxide and water;
- Vibrational transition of a molecule; Gross and specific selection rules; Fingerprint region;
- IR frequencies of functional groups; Interpreting IR spectral data.

Recap: Interpreting IR spectra

C=O is highly sensitive to its environment

EACH DIFFERENT KIND OF C=O COMES AT A DIFFERENT FREQUENCY



1810 and 1760

THESE VALUES ARE WORTH LEARNING

Factors affecting absorption frequency

Absorption frequency:

- Increases with increase in % of s character: $sp^3 < sp^2 < sp$

e.g. C-H stretching frequencies:

< 3000 cm^{-1} for sp^3, >3000 cm^{-1} for sp^2, ~3300 cm^{-1} for sp

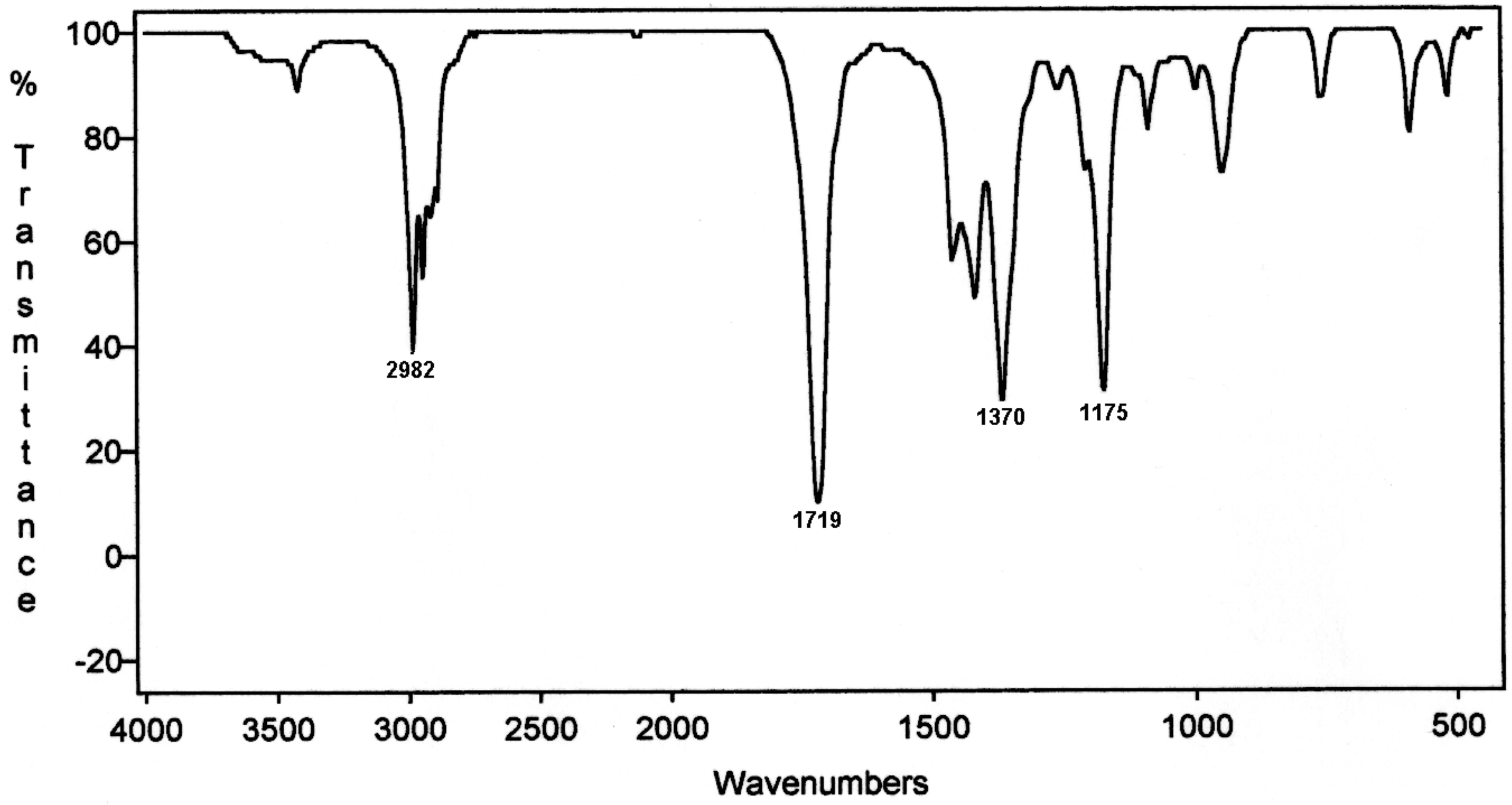
- Decreases on increase in conjugation

e.g. In aldehydes, ketones and esters, C=O stretching frequency decreases by 25~30 cm^{-1} when conjugated with C=C

- Is affected by inductive/resonance/electromeric effects
- Decreases due to hydrogen bonding

Interpreting IR spectra of 2-butanone [Self-study]

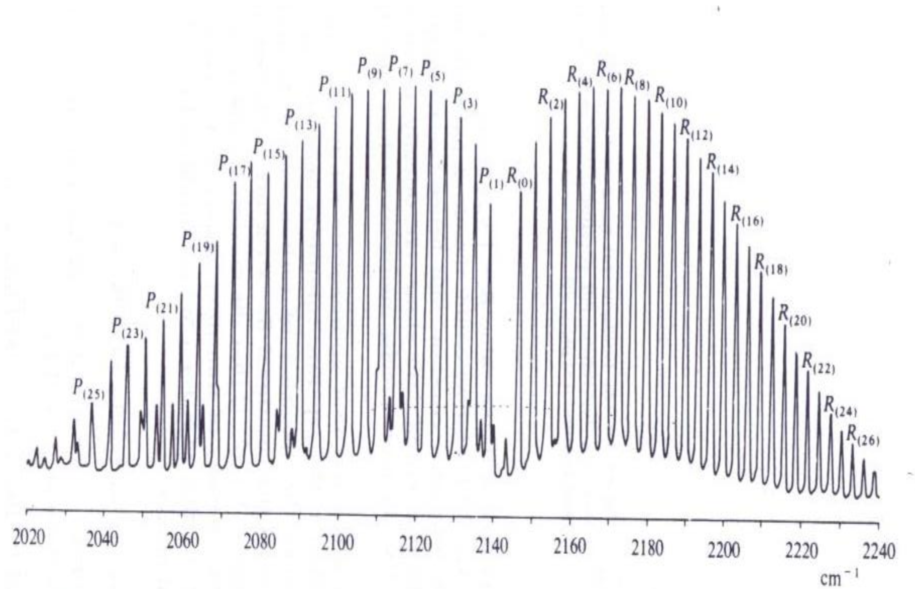
Question: Find out the overtone of the strong C=O peak that you see at 1719 cm^{-1}

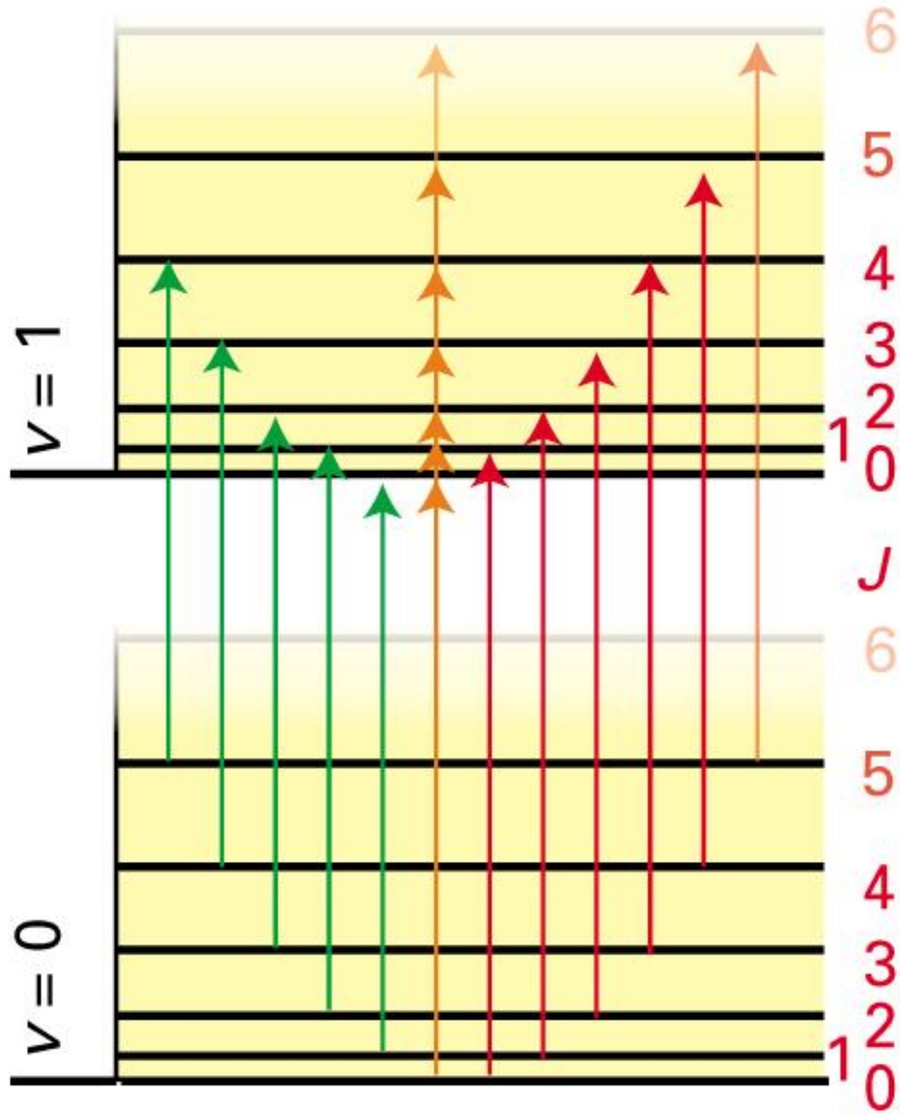


Vibration-Rotation Spectra

Vibrational transitions accompanied by rotational transitions – band structure.

Diatomic vibrating rotor (CO)





Rotational spectroscopy (Microwave spectroscopy)

Pure Rotational Spectroscopy

- Transitions between rotational energy levels are involved
- Mostly gas phase measurements; Long path lengths to achieve sufficient absorption.
- Also known as microwave spectroscopy.

First Approximation

Rigid Rotor model: No structural distortion due to rotational motion (In reality, there the centrifugal distortion due to molecular rotations)

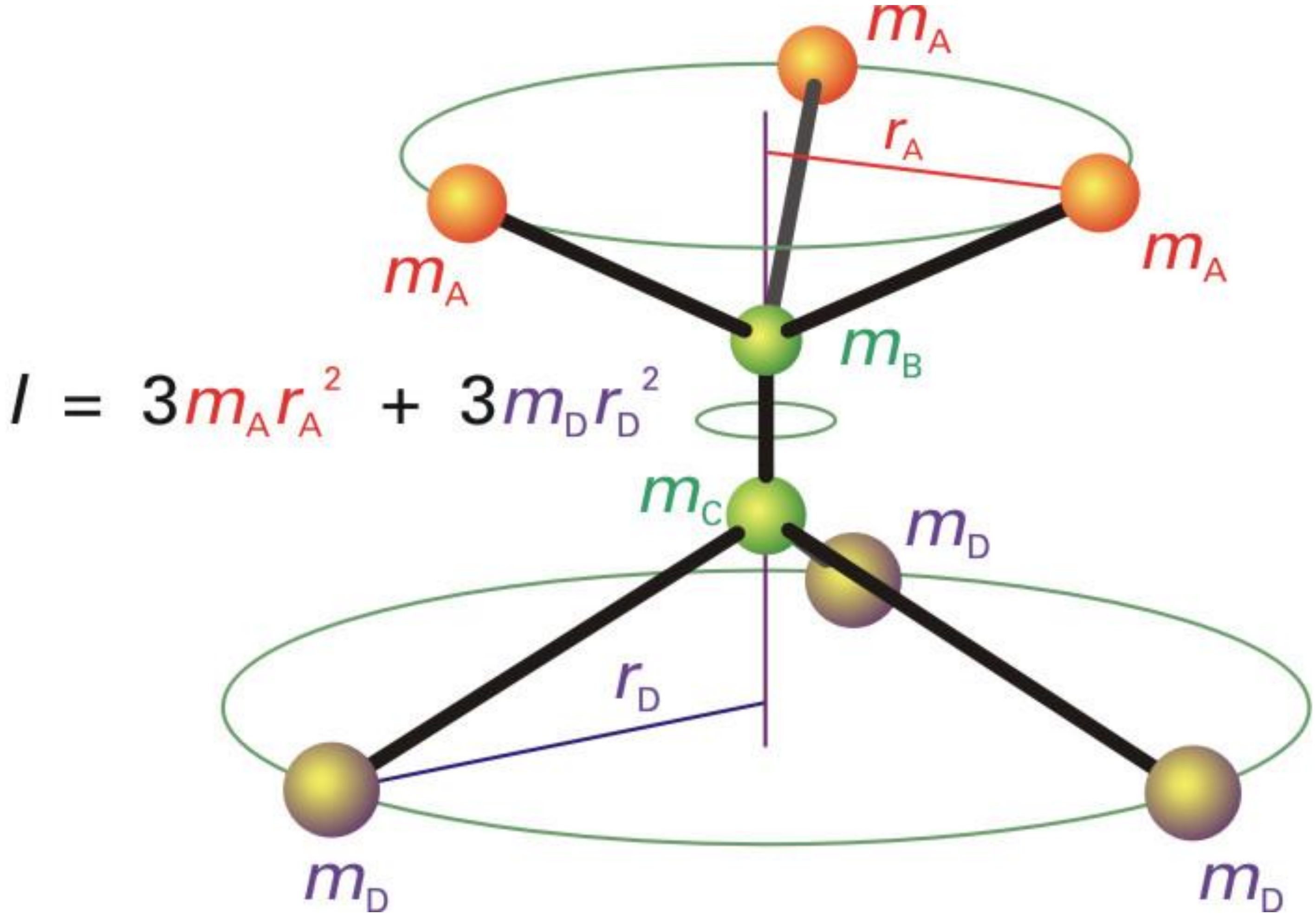
Pure Rotational Spectra of Linear Molecules

- Rotational quantum number: J=0,1,2,....
- Angular momentum = $[J(J+1)]^{1/2}\hbar$
- Rotational energy levels:
- $E_J = J(J+1)\hbar^2/2I = hB\, J(J+1)$
- $B = (\hbar/4\pi I)$ is rotational constant
- Degeneracy: 2J+1 (m_J = -J,-J+1,..,0,...,J-1,J)

Pure Rotational Spectra of Linear Molecules

- Axis of rotation – perpendicular to the molecular axis, passes through the center of mass – Application of rigid rotor
- Moment of intertia $I = \sum_k m_k r_k^2$
- For diatomic molecules,
- $I = m_1 r_1^2 + m_2 r_2^2$
- Homework: Show that this simplifies to $I = \mu r^2$, where μ is reduced mass of the molecule and r is the bond-length.

Rotational Spectroscopy

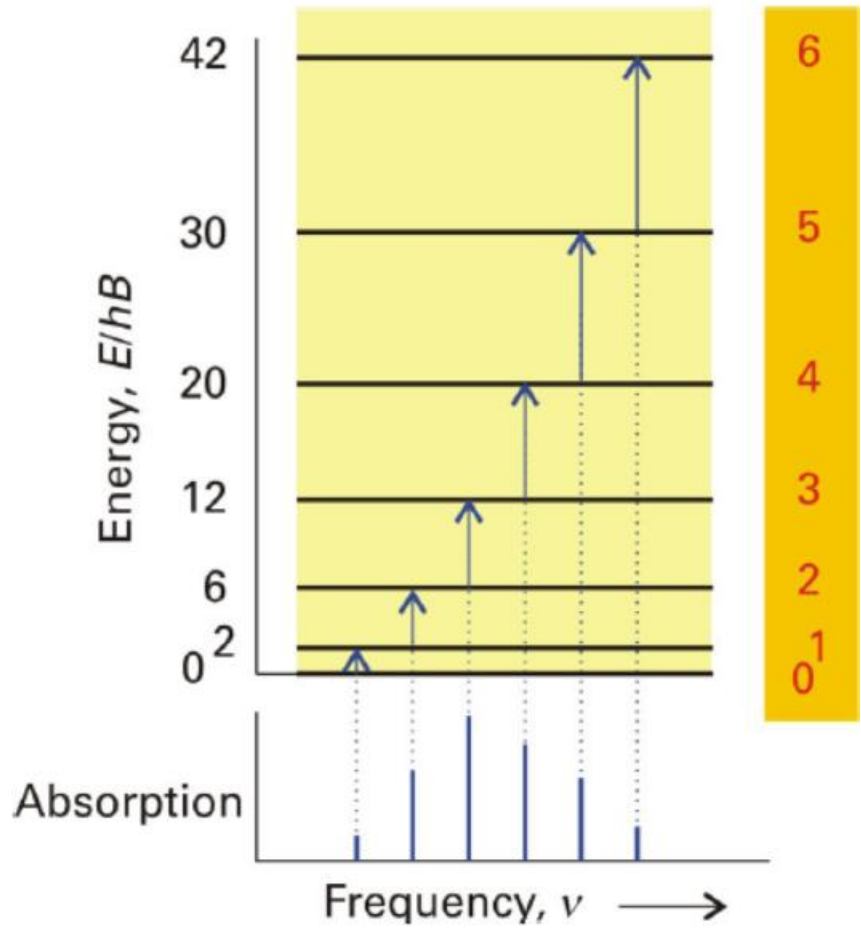


Moment of Inertia (ref. to Tab.19.1 of TB)

Pure Rotational Spectra of Linear Molecules

- $\Delta E_{J+1\leftarrow J} = 2hB(J+1)$, $J=0,1,2,\ldots$
- $\nu_{J+1\leftarrow J} = 2B(J+1)$
- ~10 GHz → λ~0.1-1 cm (microwave)

The separation between successive spectral lines is **2B** from which we can find **I**, and hence the bond length.



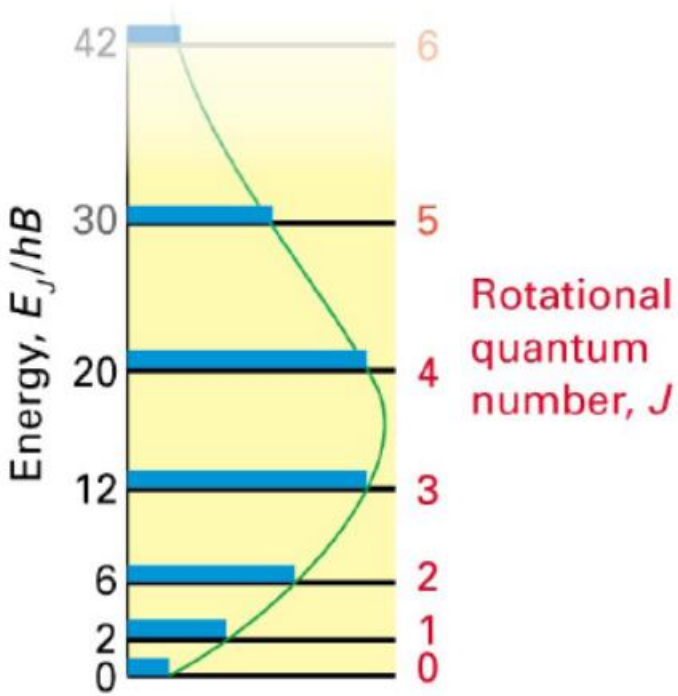
Population of rotational levels in linear molecules

- At room temperature, many rotational levels of molecules are populated as per Boltzmann distrubution.
- Degeneracy of energy levels in linear rigid rotor = *(2J+1)*
- The population of J^{th} level relative to the ground level (*J=0*) in linear rotor is given by $$\frac{P_J}{P_0} = (2J+1)e^{-hBJ(J+1)/kT}$$

The distribution passes through a maximum which can be determined by assuming P_J to be a continuous function and setting $dP_J/dJ = 0$

It follows: $$J_{max} = \sqrt{\frac{kT}{2hB}} - \frac{1}{2}$$

(Nearest integer value of *J* to be considered)



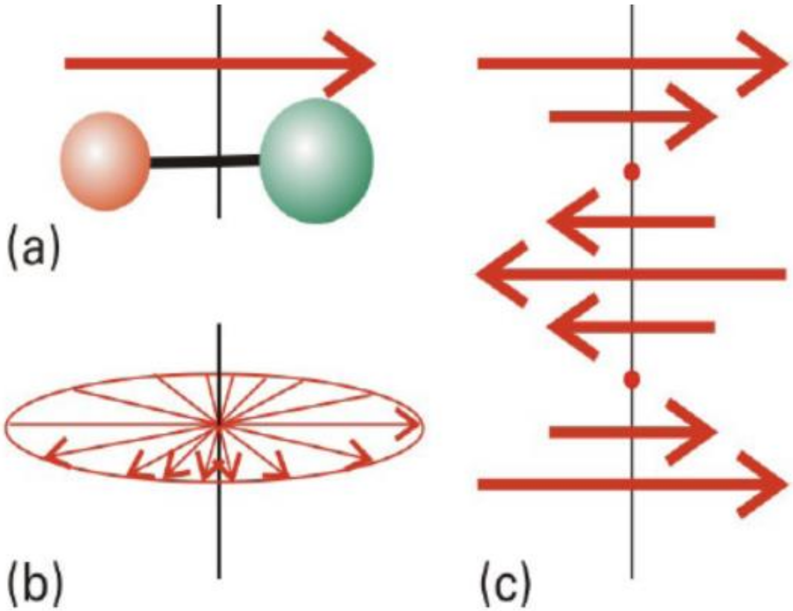
Pure Rotational Spectra of Linear Molecules

Transition dipole moment : $\int \psi_{final}\, \mu \psi_{initial} \quad d\tau$

Square of this value determines intensity. If transition dipole moment is 0 then no transition takes place.

- Gross selection rule: The molecule must posses permanent dipole moment in order to be microwave active

- Specific selection rule: $\Delta J = \pm 1$ - conservation of angular momentum upon absorption/emission of photon



•***No Rotational spectrum for CH_4, SF_6, CO_2, H_2, O_2 etc. (Rotationally inactive)***

•***Rotational spectra observed for HCl, NH_3, CH_3Cl etc.***

Rotational Spectroscopy

- ***Centrifugal distortion: (not so rigid)***
- $E_J = hBJ(J+1) - hDJ^2(J+1)^2$
- ***D is the centrifugal distortion constant, whose magnitude is related to the force constants of the bonds.***
- ***B also depends on vibrations.***

$\nu_J = 2B\,(J+1) - 4D(J+1)^3$

As the J increases the lines converge.

$\nu_J\,/\,(J+1) = 2B - 4D(J+1)^2$

Therefore, a plot of $\nu_J\,/(J+1)$ ***vs.*** $(J+1)^2$ ***will give -4D as slope and 2B as intercept.***

Raman spectroscopy

Raman spectroscopy

Original research at IACS was started in 1907 by C. V. Raman (1888-1970), then 19 years old. Raman came to Calcutta as Assistant Auditor General of Government of India and started working at IACS beyond office hours and on holidays. He very quickly published his first paper from IACS (*Nature* 24 Oct, 1907). A. L. Sircar (son of M. L. Sircar) said in 1907 that "*Raman will be the brightest ornament of IACS*." The initial experiments on Raman effect were done using sunlight as the light source and eye as detector. G. D. Birla provided Raman with the money to buy the spectrograph. The Raman Effect was discovered in 1928. For this discovery, Raman received the *1930 Nobel Prize in Physics*. Raman announced this discovery on 28th February, 1928. This day is celebrated as the *National Science Day* in India. It is said that Raman booked tickets to go to Stockholm 6 months in advance but cried during Nobel Prize ceremony*because there was no Indian Flag on his chair*! From 1917-33 Raman was a Palit Professor of Physics at the Calcutta University. But he carried out all his research at IACS. Raman discovered the small angle X-ray scattering in 1929.

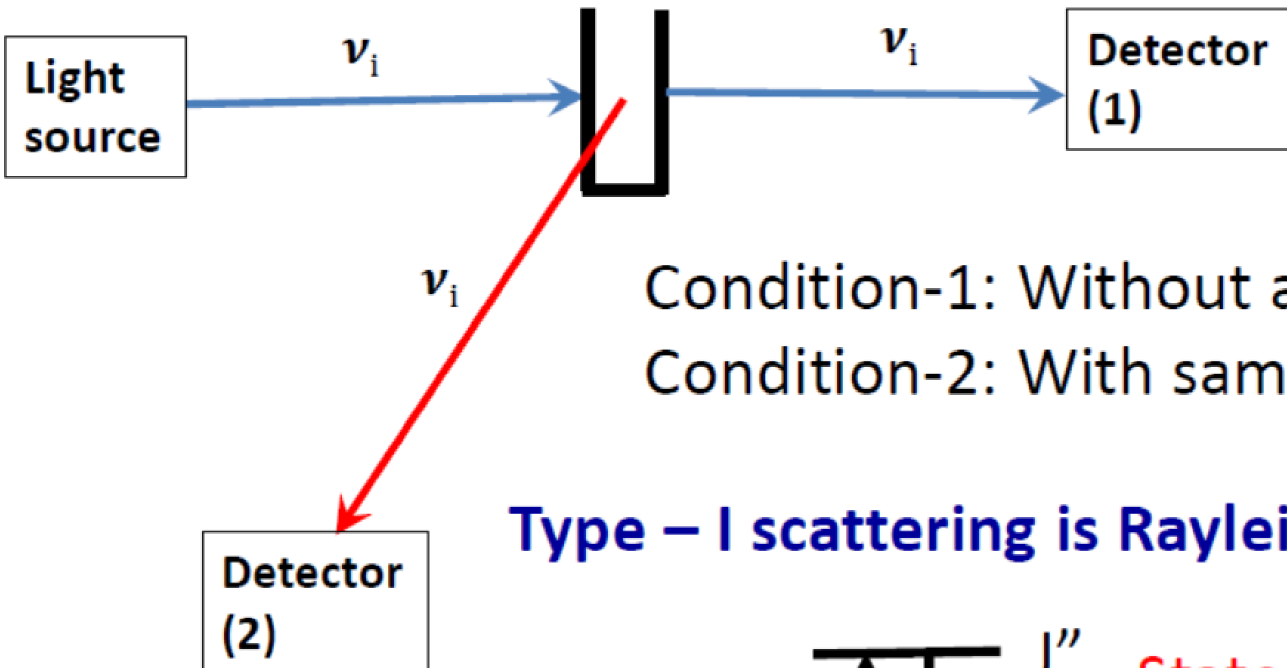
Worked at:
University of Calcutta.
Banaras Hindu University.
IISc, Bangalore.
Raman Research Institute, Bangalore.

Nobel Prize in Physics, 1930

The Nobel Prize in Physics 1930 was awarded to Sir Chandrasekhara Venkata Raman *"for his work on the scattering of light and for the discovery of the effect named after him"*.

Raman spectroscopy

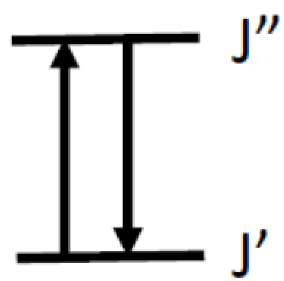
Type – I: Elastic scattering



Condition-1: Without any sample

Condition-2: With sample

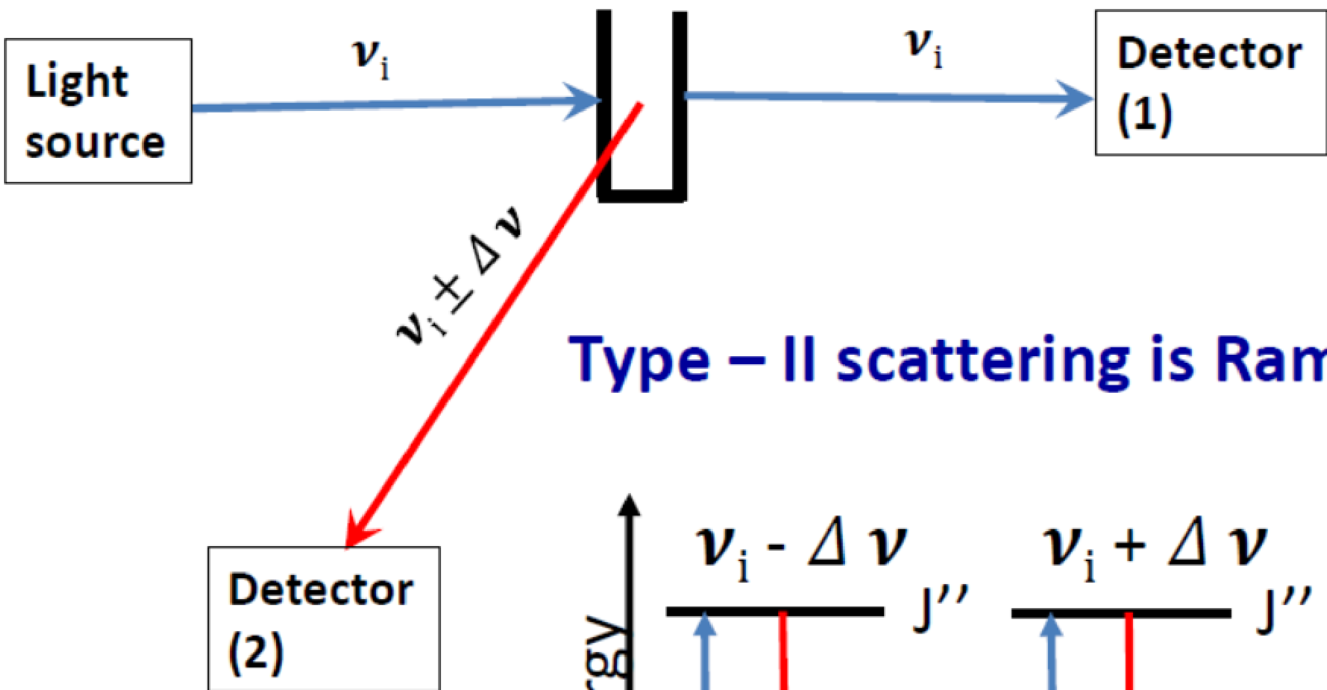
Type – I scattering is Rayleigh scattering



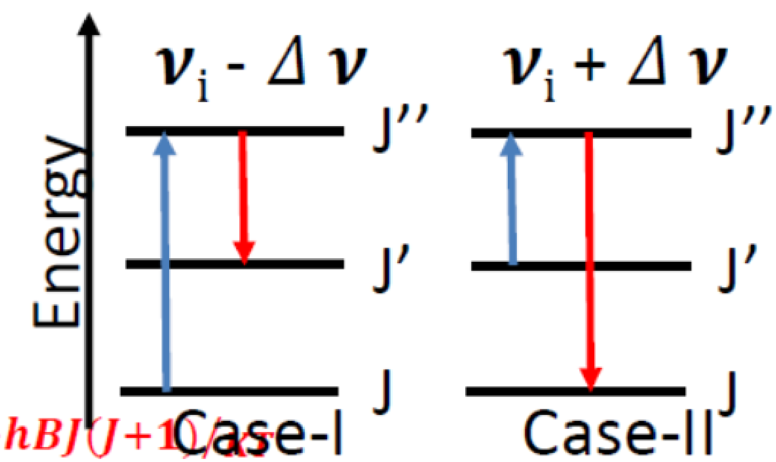
State of the system remains unchanged

Raman spectroscopy

Type – II: Inelastic scattering



Type – II scattering is Raman scattering



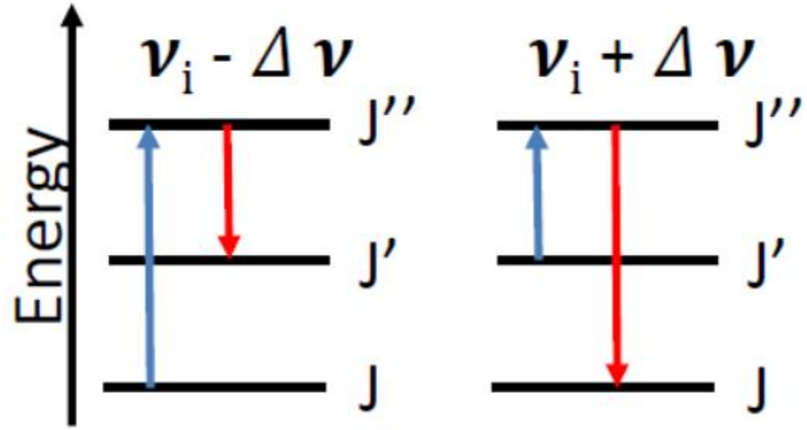
$$\frac{P_J}{P_0} = 7.8$$

$$P_J = P_0\,(2J+1)\,e^{-hBJ(J+1)/kT}$$

Raman spectroscopy

Case-I | **Case-II**

Case-I: Molecule gains energy, ΔE; State changes from J to J'.



Case-II: Molecule release energy, ΔE; State changes from J'' to J'.

$$\nu' = \nu_i - \Delta\nu$$

$$\Rightarrow \nu_S = \nu_i - \frac{\Delta E}{h}$$

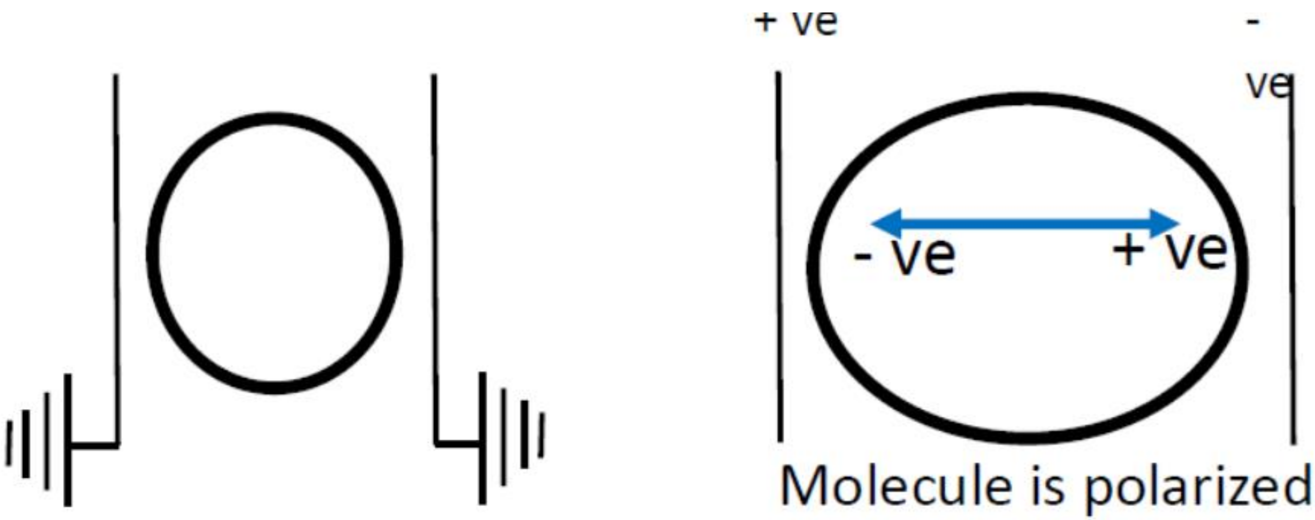
Stokes' radiation

$$\nu'' = \nu_i + \Delta\nu$$

$$\Rightarrow \nu_{as} = \nu_i + \frac{\Delta E}{h}$$

Anti-Stokes' radiation

Raman spectroscopy



This separation of charge centers **causes an induced electric dipole moment**

Polarization leads to dipole moment in the molecule – induced dipole moment

Induced dipole: $\mu = \alpha E; E: applied\ field$

α is a parameter – the ease with which a molecule can be distorted - POLARIZABILITY

Raman spectroscopy

Gross Selection Rule

Some internal motion of the molecule, such as, rotation and/or vibration must change the polarizability, i.e., $\frac{d\alpha}{dq} \neq 0$

q represents the coordinate in terms of the internal motion

If molecular rotation/vibration causes some change in molecular polarizability then *those rotation/vibration must be Raman Active*

All diatomic molecules are Raman active.
Spherical rotors such as CH_4, SF_6 are inactive.

Conditions imposed on rotation model we have used:

Molecules must be non-rigid and hence polarizable

Raman spectroscopy

- Based on scattering of the radiations incident on the sample (molecules).
- During the scattering, the energy may be exchanged between the sample and the radiation.
- Lasers commonly used as source in Raman – frequency shifts being small, and scattering intensity being low.
- Raman spectroscopy is observable if the polarizability of the molecule changes in a given transition, due to scattering.
- Stokes radiation – scattered radiation of lower frequency than incident radiation, energy transfer to molecule.
- Anti-Stokes radiation – scattered radiation of higher frequency than incident radiation, energy transfer from molecule.
- Rayleigh radiation – no energy transfer, no change in frequency – this is the most intense line in the Raman spectra of molecules

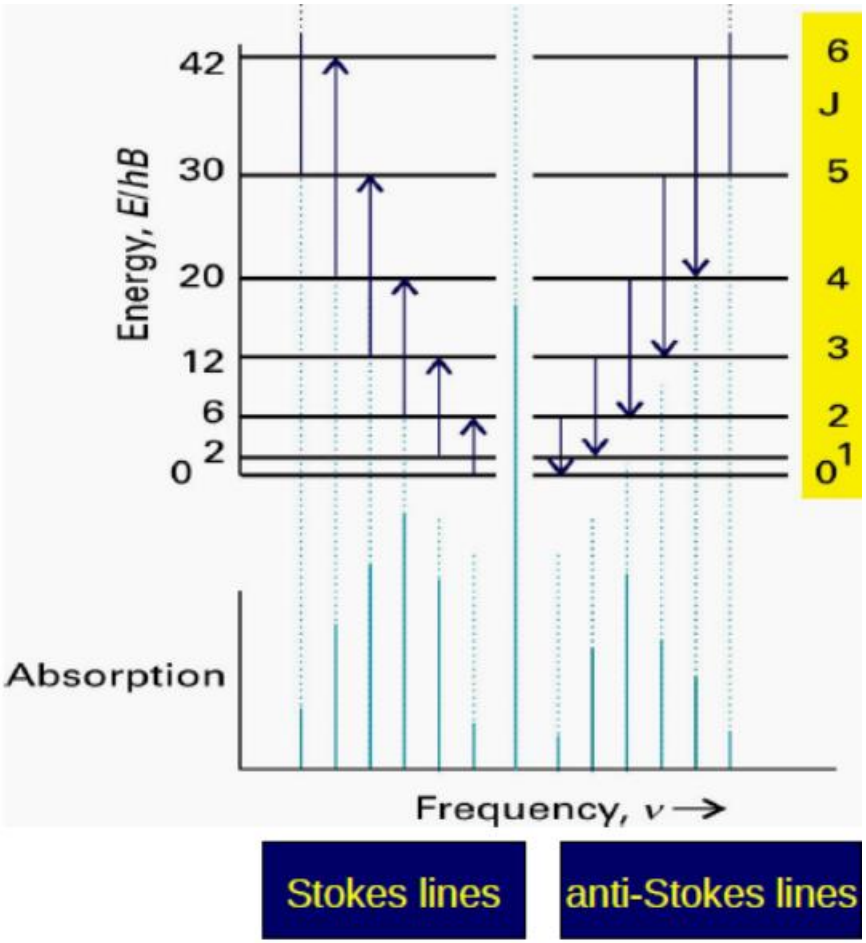
Rotational Raman spectroscopy

- The spacing in rotational energy level increases with J
- The intensities in the Stokes and the anti-Stokes lines go through maxima corresponding to J_{max}
- **Gross selection rule:** Polarizability must be anisotropic. All diatomics are active. Spherical rotors like CH_4, SF_6 are inactive.
- **Specific selection rule** (linear molecules): $\Delta J = +2$ (for Stokes lines); $\Delta J = -2$ (anti-Stokes lines)

$$\Delta E = hB[(J+2)(J+3) - J(J+1)]$$

$$= 2hB(2J+3)$$

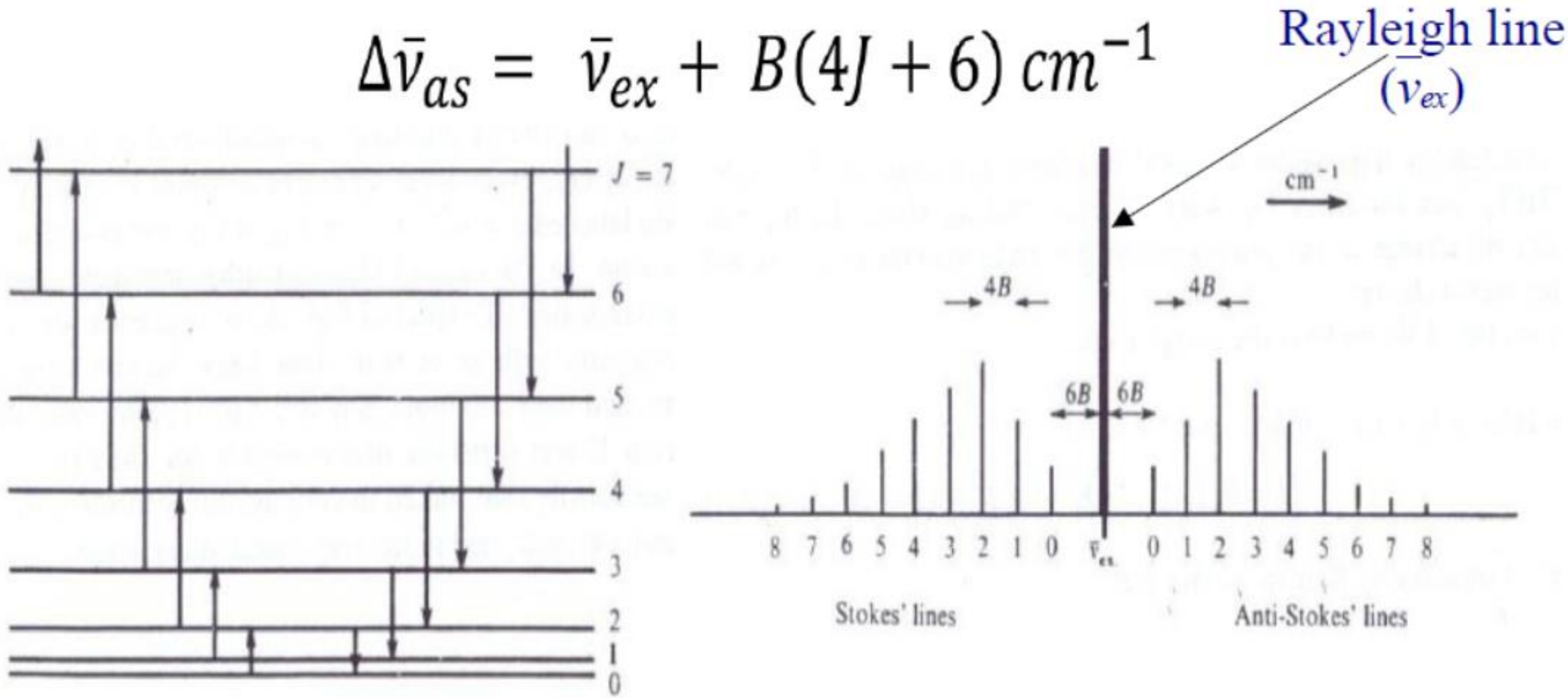
- Stokes lines at 6B, 10B, 14B,… *lower than* incident and anti-Stokes at 6B, 10B, 14B,… *higher than* incident frequency.



Rotational Raman spectroscopy

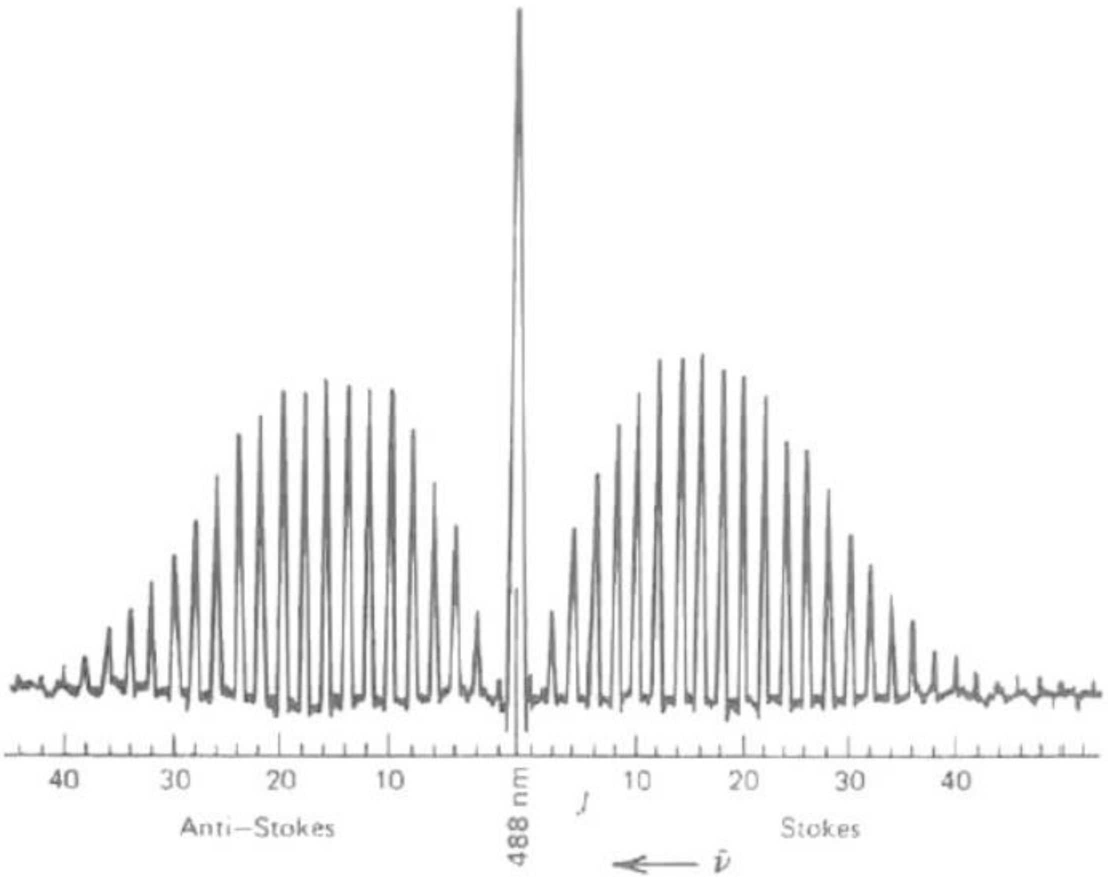
$$\Delta\bar{\nu}_s = \bar{\nu}_{ex} - B(4J+6)\ cm^{-1}$$

$$\Delta\bar{\nu}_{as} = \bar{\nu}_{ex} + B(4J+6)\ cm^{-1}$$



Stokes lines at 6B, 10B, 14B,… lower than incident and Anti-Stokes at 6B, 10B, 14B,… higher than incident frequency.

Rotational Raman spectroscopy



Rotational Raman spectrum of carbon dioxide

Applications

Homonuclear diatomics:

H_2, O_2, N_2: Rotationally inactive (non-polar)

Active in rotational Raman

Rotational Raman spectroscopy provides information about the geometric parameters

Vibrational Raman spectroscopy

innovate achieve lead

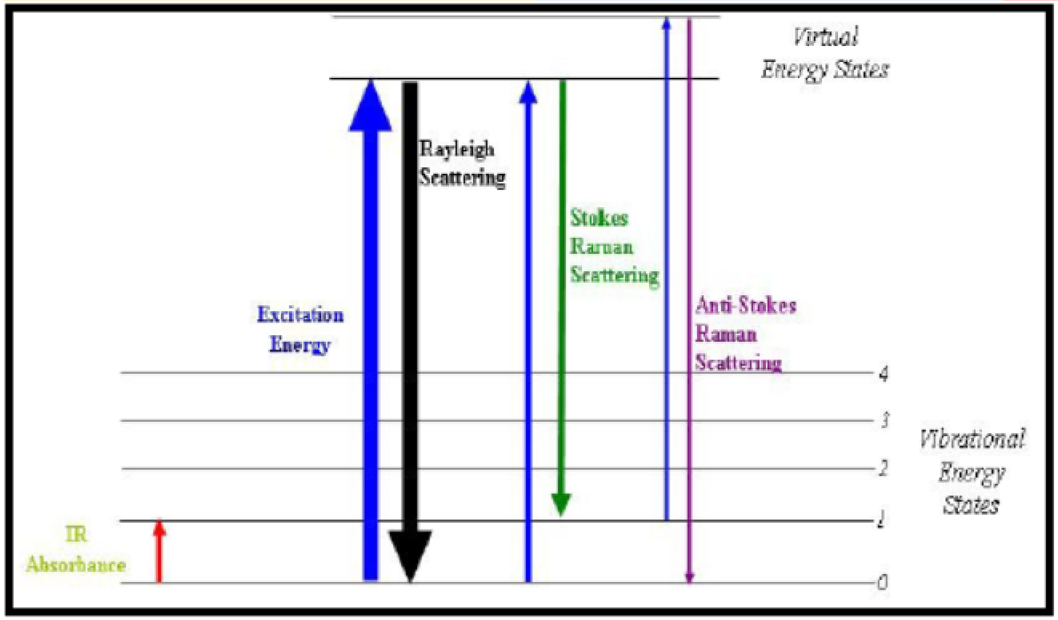
Vibrational Raman spectra:

- ***Normal modes can be either IR or Raman active (or both);***
- ***In molecules having centre of inversion, no mode can be both IR and Raman active (e.g. CO_2 where symmetric stretching is only Raman active and other modes are only IR active).***

In other words,

In case of molecules with center of inversion, the Raman active vibrational modes are IR inactive and IR active vibrational modes are Raman inactive. However, it is possible that a normal mode may be both IR and Raman inactive.

Vibrational Raman Spectroscopy



The spacing in vibrational energy levels remains almost constant (slightly decreases with increase in v, due to anharmonicity)

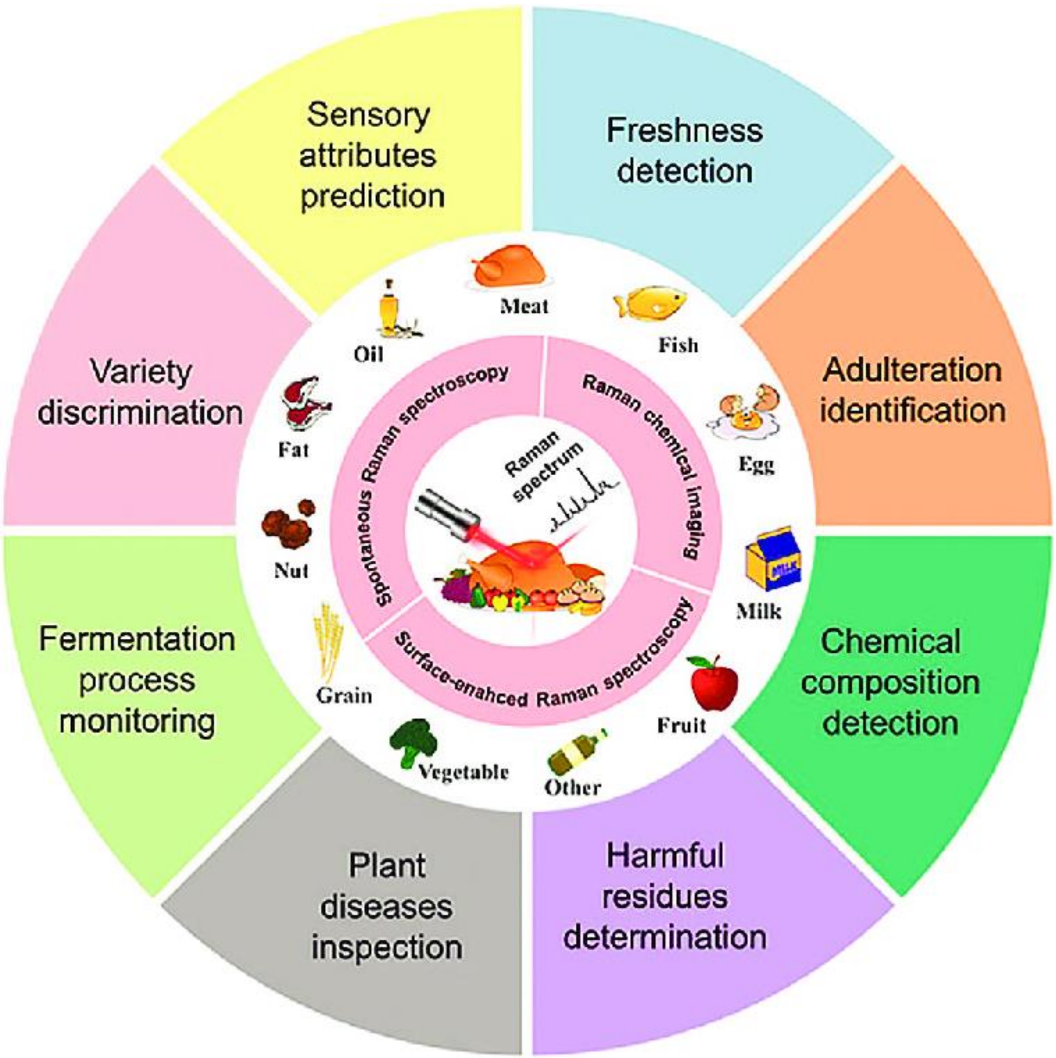
Gross selection rule: Polarizability must change during transition

Specific Selection rule (for linear molecules):
$\Delta v=+1$ (for Stokes lines); $\Delta v=-1$ (for anti-Stokes lines)
(Note that this is same as in case of pure vibrational (IR) spectroscopy)

At room temperature, most of the molecules are in ground vibrational state. Therefore, the intensity of first Stokes line is predominantly high compared to other Stokes lines and all anti-Stokes lines in Vibrational Raman spectra.

Applications of Raman Spectroscopy

[Not in syllabus]



3D Raman imaging of squashed banana pulp